PROJECT 11 · AUTOMATIC CLOTHING CATEGORY RECOGNITION

Phase 2: Teaching a CNN to See Clothes



From handcrafted features to a calibrated, reliable CNN

95.20%	480	1.28%	98.56%
final test accuracy	errors (SVM: 870)	calibration error	when confident

Team T002 · Newton School of Technology · Deep Learning

MD. Sajjan, Sibtain Ahmed Qureshi, Syed Darain Qamar, Antik Mandal, Abuzar Haideri

THE STORY IN ONE SLIDE

From handcrafted features to learned features



Phase 1

Classical ML + HOG features

Best: SVM-RBF 91.30%



The ceiling

74% of errors in 4 upper-body classes

Shirt recall only 75%



Phase 2

CNNs learn features from raw pixels

Best: CNN3Block-GAP 93.09%



Phase 2 final

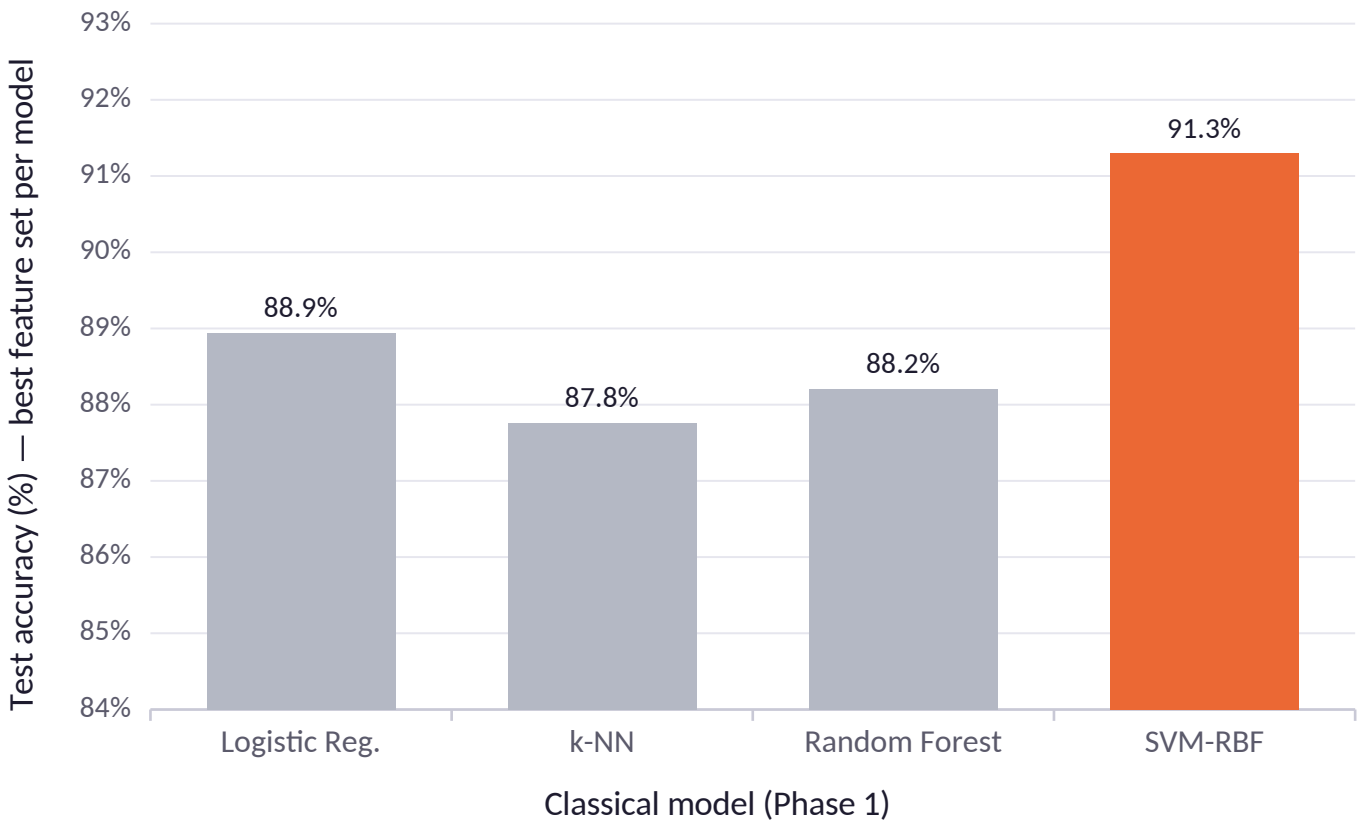
Ensemble + calibration + abstain

Final: 95.20%

Takeaway Phase 2 broke the 91.3% ceiling, explained why, and finished with a model whose confidence can be trusted.

PHASE 1 RECAP

Classical ML hit a ceiling at 91.3%



91.30%
best classical accuracy (SVM-RBF on HOG)

74%
of the SVM's errors are T-shirt, Pullover, Coat or Shirt

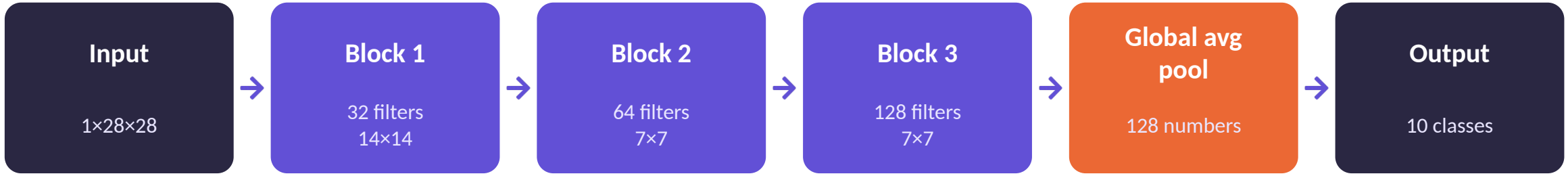
75.0%
Shirt recall — 1 in 4 shirts mislabelled

Takeaway Handcrafted HOG features cannot see the subtle collar and hem details that separate a Shirt from a T-shirt.

ARCHITECTURES

Four CNNs in a 2×2 experiment

Our best network — CNN3Block-GAP



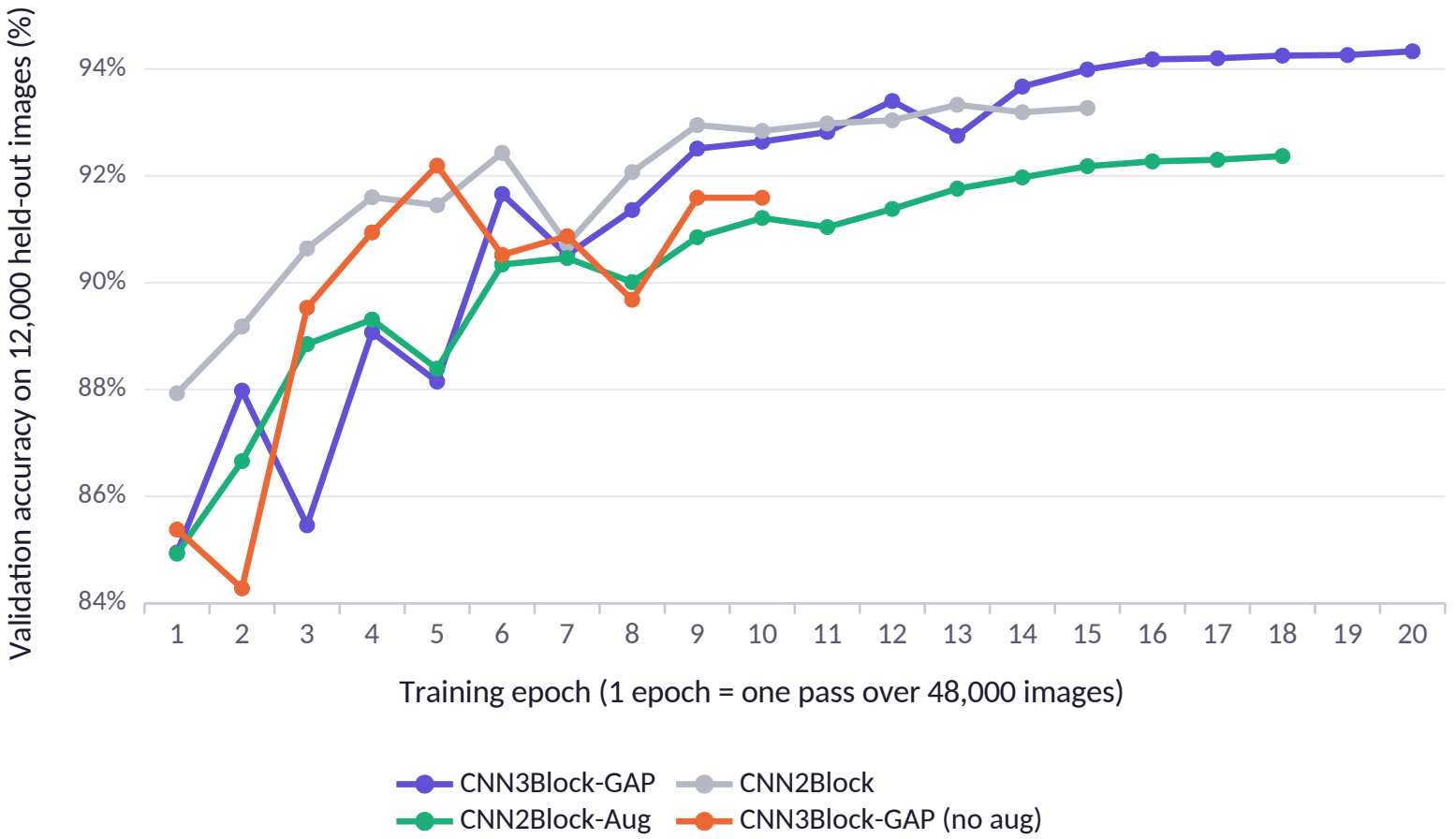
Each block = two 3×3 convolutions + batch-norm + ReLU. Global average pooling replaces the big flatten layer → fewer weights, less overfitting.

	No augmentation	Crop + flip augmentation
CNN2Block (flatten, 422k params)	**CNN2Block**	**CNN2Block-Aug**
CNN3Block-GAP (305k params)	**CNN3Block-GAP-NoAug (new run)**	**CNN3Block-GAP ★ best**

Takeaway Same 48k / 12k / 10k split as Phase 1 · every choice made on validation · test set used once.

RESULTS · TRAINING

The GAP network was still improving when training stopped



Still climbing

CNN3Block-GAP's best epoch was its last (20). More epochs will likely help.

Under-trained

CNN2Block-Aug scores higher on validation than on training — it needed more epochs.

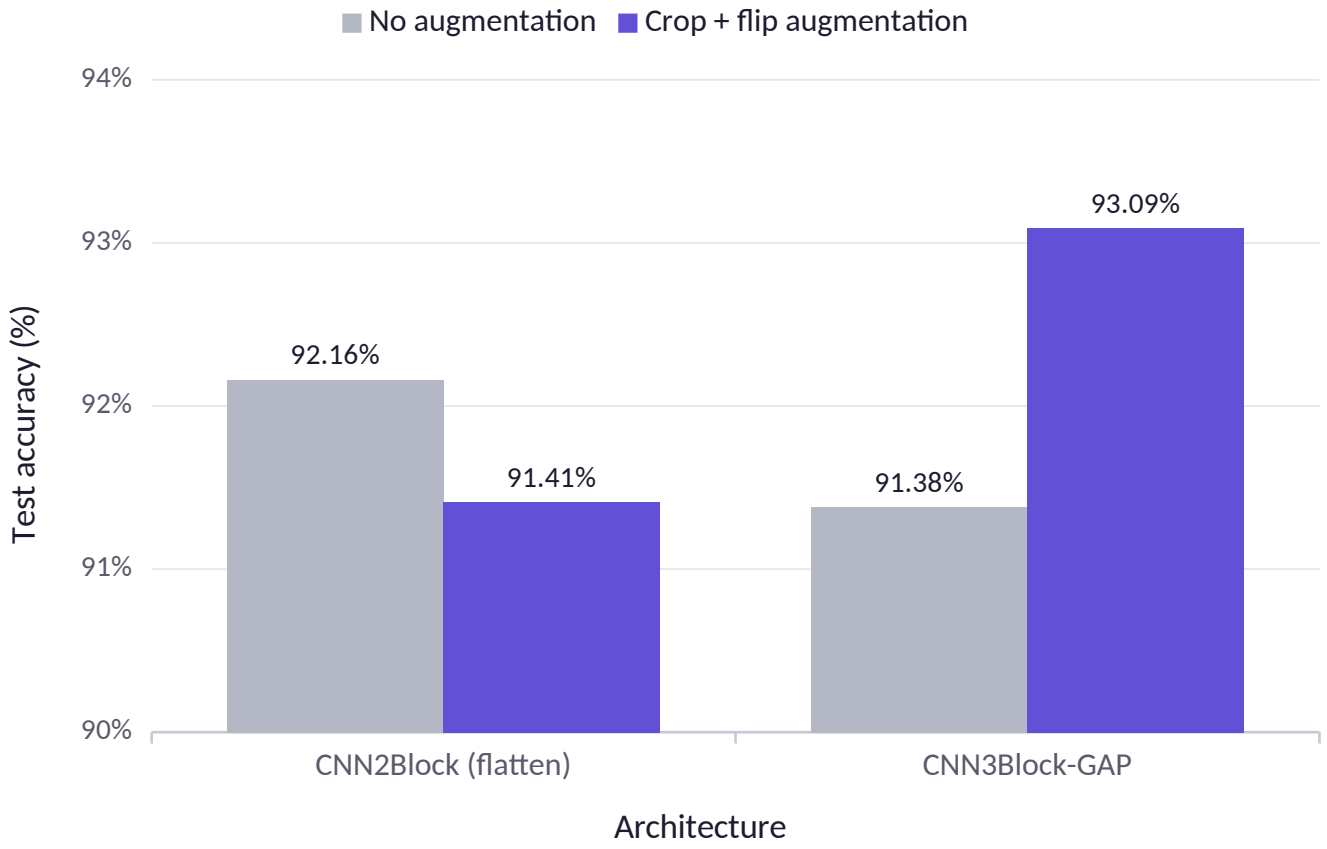
Overfits without augmentation

The GAP net without augmentation peaked at epoch 5 and was stopped at 10.

Takeaway Training budgets — not model limits — ended the best runs. So the final models get 40 epochs.

RESULTS · WHAT CAUSED THE GAIN?

Depth + GAP only pays off together with augmentation



How to read the 2×2

Augmentation on the shallow net: 92.16% → 91.41% (under-trained in 18 epochs)

Depth + GAP alone: 91.38% — overfits after 5 epochs

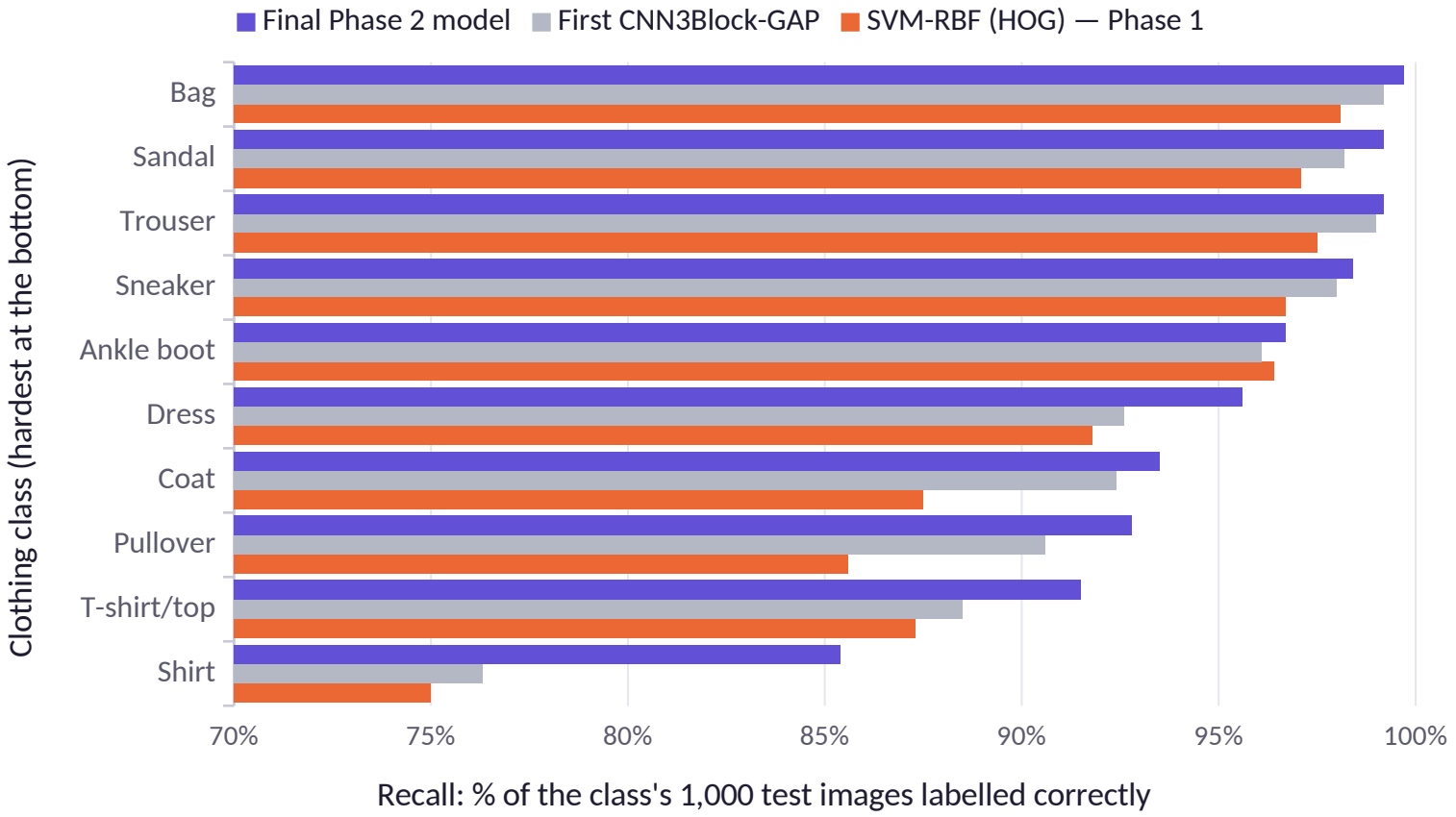
Both together: 93.09% — augmentation adds +1.71 pts to the GAP net

McNemar between the two GAP runs: $p = 1.0 \times 10^{-12}$

Takeaway Neither ingredient works alone: the deeper network needs augmentation to use its extra capacity.

ERROR ANALYSIS · PER CLASS

Every class improved — Shirt gained the most (+10.4 pts)



Shirt **+10.4 pts**

75.0% → 85.4% — still the hardest

Pullover **+7.2 pts**

85.6% → 92.8%

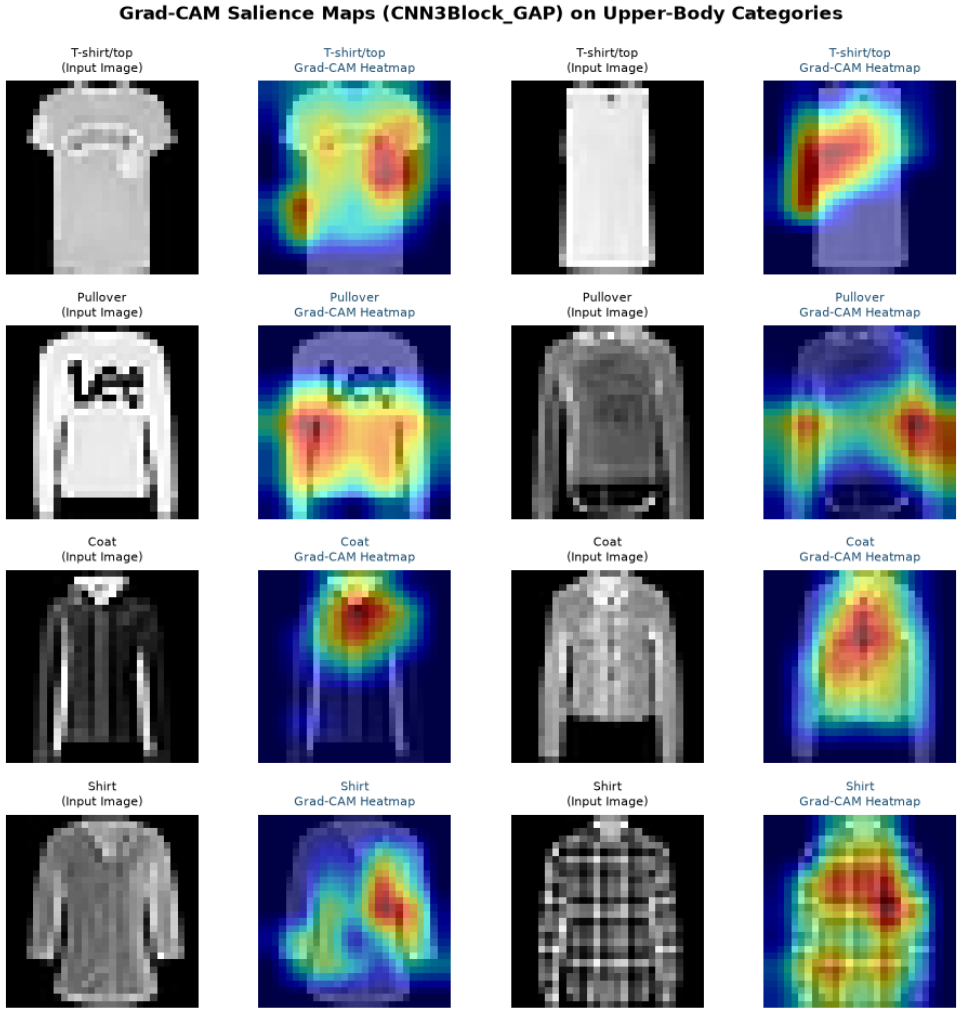
Coat **+6.0 pts**

87.5% → 93.5%

Takeaway The residual ensemble finally closes most of the Shirt gap that handcrafted features and the first CNN left open.

EXPLAINABILITY

Grad-CAM: where does the CNN look?



Red = image regions that pushed the network toward its answer.

Ignores the background

Heat sits on the garment, never on the black border.

Pullover: cuffs & waistband

Ribbed edges are a learned cue HOG could not isolate.

Some maps are diffuse

Checked Shirt and printed T-shirt light up everywhere — consistent with Shirt still being hard.

Needs measurement

8 hand-picked images are evidence, not proof — listed as a limitation.

FINAL MODEL

From a good model to a reliable one

Longer + repeated

40-epoch budget and 3 random seeds per model

results stable across seeds

Residual network

ResNetSmall with Random Erasing and label smoothing

deeper without vanishing gradients

Ensemble + TTA

Average several nets and a mirrored copy of each image

fewer random mistakes

Trustworthy output

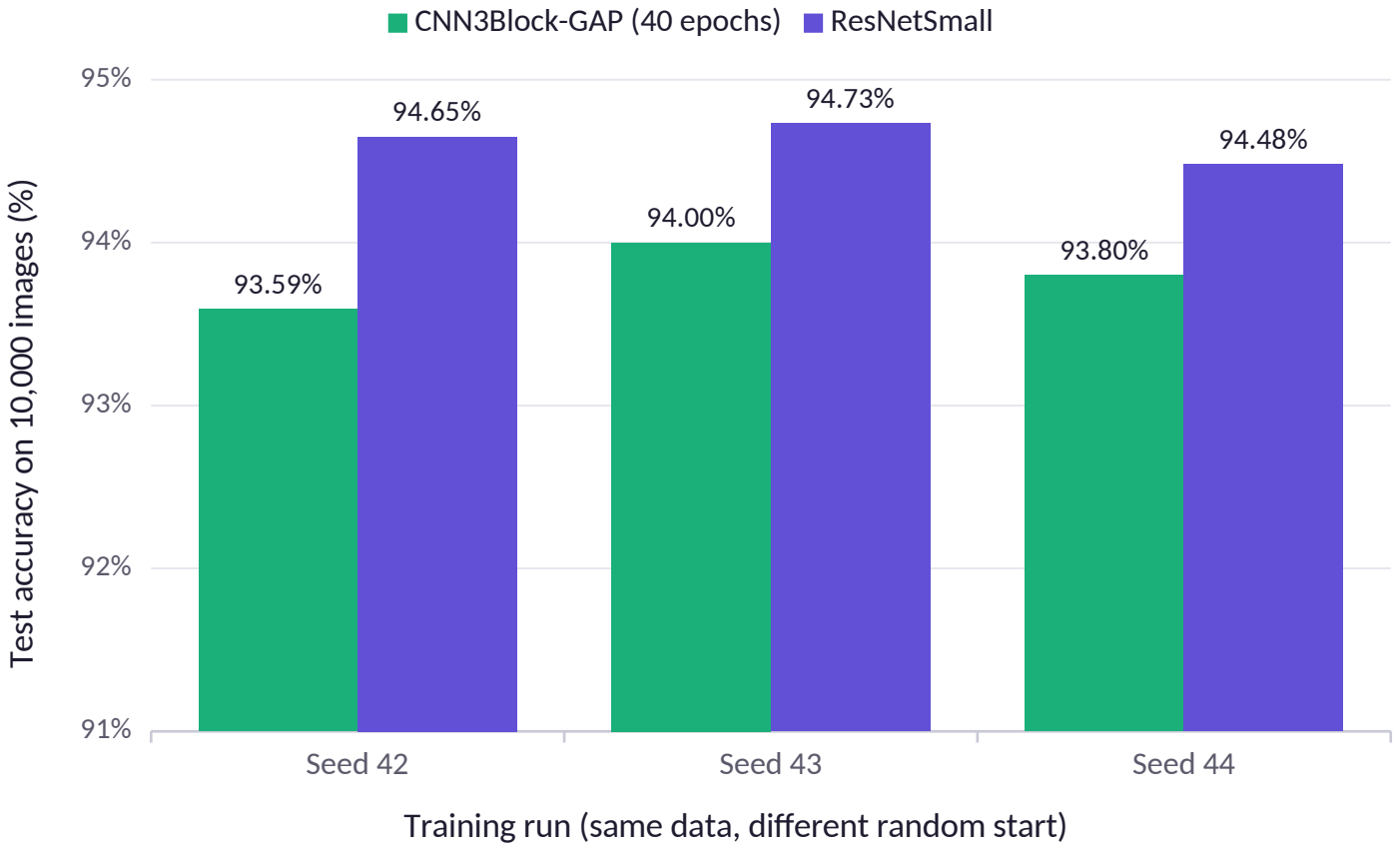
Temperature-scaled confidence, abstain option, Shirt-aware rule

know when to ask a human

Takeaway Every choice (which ensemble, temperature, Shirt rule, abstain threshold) was made on validation data only.

FINAL MODEL · ROBUSTNESS

Results hold across three random seeds



CNN3Block-GAP (40 epochs)

93.80% ± 0.21

mean ± standard deviation over seeds

ResNetSmall

94.62% ± 0.13

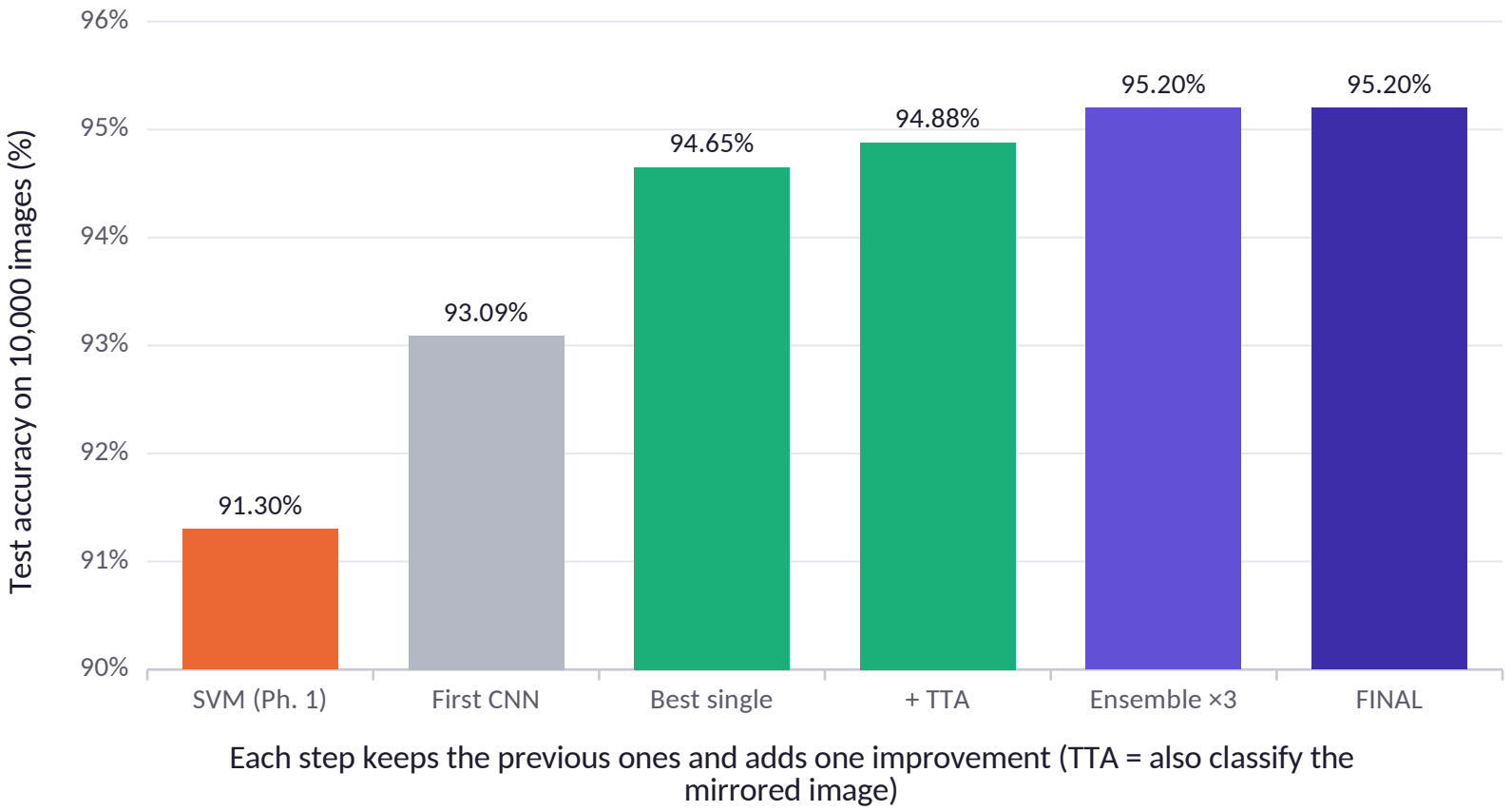
mean ± standard deviation over seeds

Every run beats the Phase 1 SVM (91.30%) and the spread between seeds is small — the improvement is not luck.

Takeaway With 40 epochs the GAP network itself improves over the first 20-epoch run — it really was under-trained.

FINAL MODEL · STEP BY STEP

How each improvement adds accuracy



Each step keeps the previous ones and adds one improvement (TTA = also classify the mirrored image)

95.20%

final test accuracy

480

errors of 10,000 (SVM: 870)

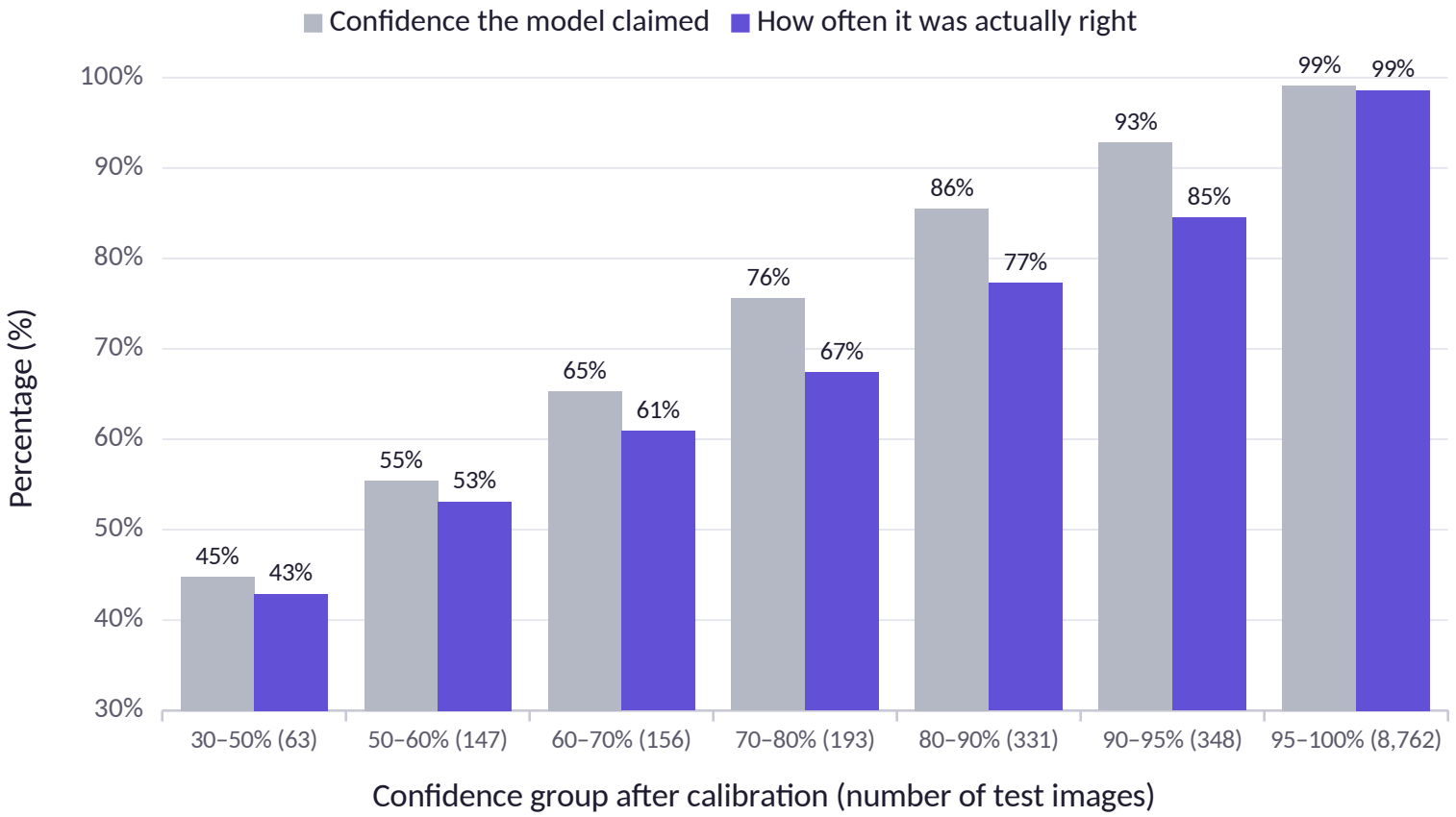
94.78%–95.62%

95% confidence interval

Takeaway Final model vs Phase 1 SVM: McNemar p = 6.7e-50; vs first CNN: p = 2.9e-27.

FINAL MODEL · CALIBRATION

Calibrated confidence: error 7.6% → 1.3%



7.64%

calibration error before (ECE)

1.28%

calibration error after temperature scaling

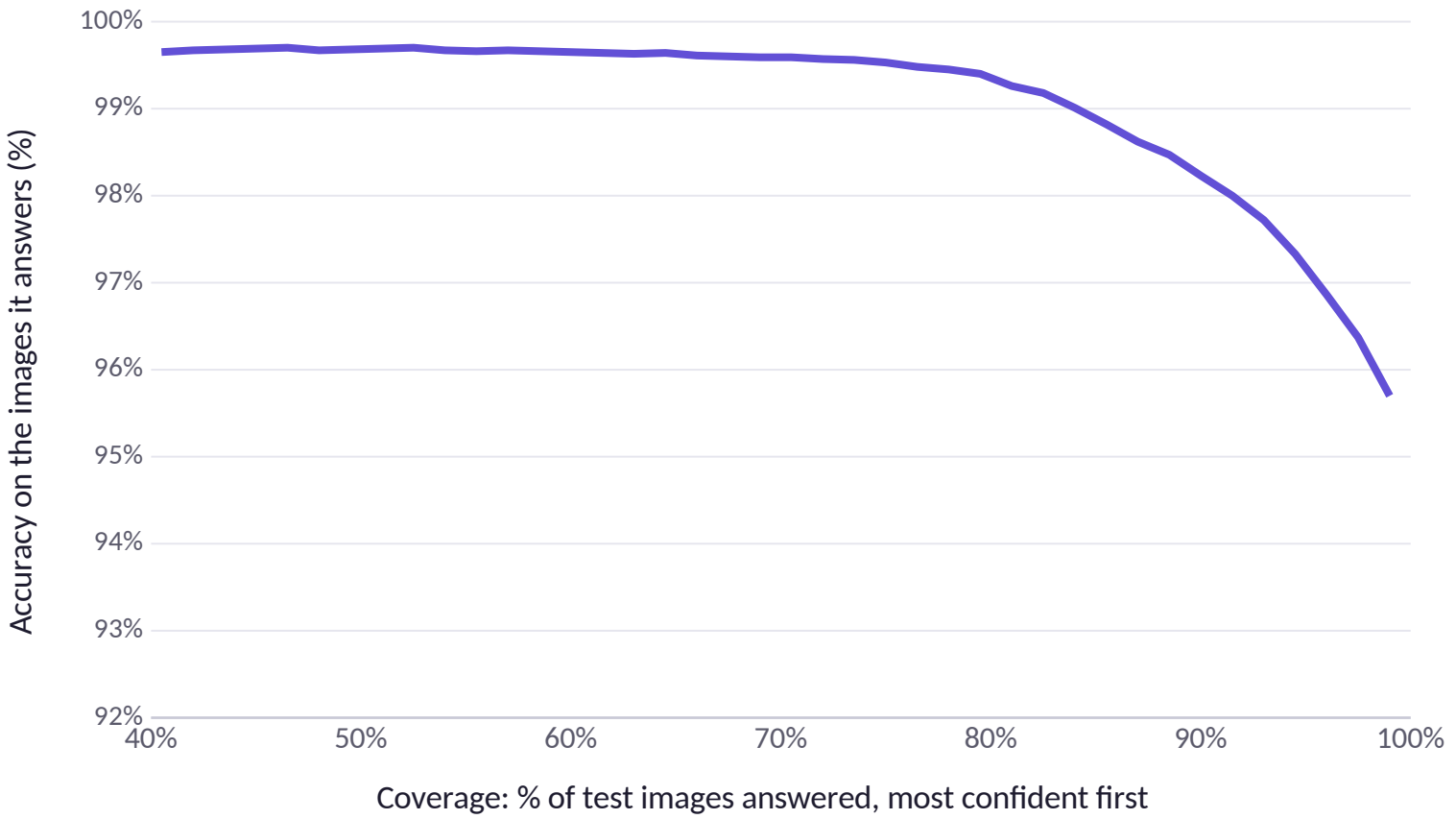
T = 0.65

temperature fitted on validation data

Takeaway 88% of images sit in the top group, where claimed and actual agree; the small middle groups stay a bit over-confident.

FINAL MODEL · KNOWING WHEN TO ASK

Let the model say “not sure” on the hardest images



87.7%

of images answered automatically

98.56%

accuracy on those images

26%

of the images sent to a human are Shirts

Takeaway Threshold chosen on validation for 99% accuracy: below 95% confidence, a human checks the image.

FINAL MODEL · SCORECARD

Phase 1 vs first CNN vs final Phase 2 model

Measure	SVM-RBF (HOG) · Phase 1	CNN3Block-GAP · first	Final Phase 2 model
Test accuracy	91.30%	93.09%	**95.20%**
Errors (of 10,000)	870	691	**480**
Shirt recall	75.0%	76.3%	**85.4%**
Calibration error (ECE)	n/a	not measured	**1.28%**
Can abstain when unsure	no	no	**yes · 98.56% on 87.7%**
Stored numbers	18.1M	0.30M	2.09M
CPU time per image	5.36 ms	0.68 ms	11.3 ms

Takeaway Most accurate and the only model with calibrated confidence — at 11.3 ms per image, still real-time.

HONEST LIMITS

What this project does not show

Shirt is still the hardest class

Even the final model misses some shirts; the abstain option routes most of them to a human.

Grad-CAM is qualitative

Heat-maps on hand-picked images are evidence, not a measured explanation.

Clean, centred thumbnails

Real shop photos (backgrounds, angles, colour) were not tested; the demo's upload is only indicative.

Small hyper-parameter search

Learning rate, width and augmentation strength were not tuned exhaustively.

Takeaway Naming the limits is part of the result: every claim in this deck is backed by a measured number.

THREE THINGS TO REMEMBER

1

Learned features beat handcrafted ones

95.20% vs 91.30% for the best Phase 1 model — 390 fewer errors.

2

We know why it works

Depth + GAP only pay off with augmentation; results hold over 3 seeds.

3

The final model knows when it is unsure

Calibrated (ECE 1.28%) and 98.56% accurate on the 87.7% of images it answers.

Thank you — questions?

Live demo: `streamlit run app/streamlit_app.py`